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UART API

Welcome to Arylic's UART API developer documentation.

These messages are mainly defined to get current system status and control the device via the UART pins on some DIY boards of Arylic, eg: Up2Stream PRO. And the product models MA400/HA400 which designed with RS232 serial port, support to control with this API.

Basic Rules

UART parameters

```
115200/8/n/1  
no flow control
```

- messages are defined in 3 characters, and will use `:` to separate the different part.
- messages sent over UART need to be terminated with `;`
- messages might be received without query when state changed.
- Content between `{ }` is variable name, you need to replace with the real content, and `{ }` itself is not meant to be sent.
- Content between `[ ]` means optional, and `[ ]` itself is not meant to be sent.
- normally, messages sent by host without param means to query current state or direct control
- messages sent with param means to control or change state.
- messages received normally with param indicating current state.

Evaluate with TCP

Most of these messages could be sent/received via TCP API, so you could also use this API with network. Normally, this is supported by the latest BP10XX based platform, from Up2Stream MINI/PRO/AMP V3 and later. You could check the silkprint on MCU to verify, it should read BP1048 or BP1064.

set volume to 50 via TCP

MCU+PAS+RAK0IT:VOL:50&

You could evaluate these commands on windows with this tool, and connect the IP of target device and send command or receive the responses. You need to pack the raw UART message into this structure: `MCU+PAS+RAK0IT:{uart_message}&`

State And Control

Device

SUMMARY

Command	Ver	Response	Ctrl
STA	3	STA:{states}	
SYS:{cmd}	3		*
WWW	3	WWW:{onoff}	
NAM[:{hextext}]	3	NAM:{hextext}	*
ETH	3	ETH:{onoff}	
WIF	3	WIF:{onoff}	
WRS	3		*
WSS	6	WSS:{rssi}	
BSS	6	BSS:{rssi}	
IPA	6	IPA:{ip}	
TME	6	TME:{time}	
COE[:{onoff}]	8	COE:{onoff}	*
COD[:{pin}]	8	COD:{pin}	*

STA response sample

NET,0,33,-2,0,1,1,1,1,0

STA

Device status summary, and the response message `{states}` will consist with: current source,mute,volume,treble,bass,net,internet,playing,led,upgrading.

reset factory

SYS:RESET

SYS:{CMD}

system operations

{cmd}	description
REBOOT	reboot device
STANDBY	enter standby. Not well considered, and can't wake up via UART for some models.
RESET	reset factory

message received when device connected to internet

WWW:1

WWW

get internet status, message would be sent when state changed

change device name to Backyard

NAM:4261636B79617264

NAM[:{HEXTEXT}]

set or get device name, and the string is encoded with hex value in UTF-8 encoding. eg: 4261636B79617264 for Backyard

message received when device connected to ethernet

ETH:1

ETH

get ethernet status, message would be sent when state changed

message received when device connected to WIFI AP

WIF:1

WIF

get wifi status, wether connected to AP. message would be sent when state changed

WRS

trigger the WIFI setup, device will enter WIFI configuration mode and waiting for setup in APP

message received when query WIFI signal strength

WSS:-49

WSS

get WIFI signal strength, host need to query this value proactively.

BSS

get Bluetooth signal strength, not working in all models and host need to query this value proactively.

message received when query IP

```
IPA:192.168.0.105
```

IPA

get device IP address, message would be sent when state changed.

message received when query current time

```
TME:2024-06-11 09:14:00 (+8)
```

TME

get local time, host need to query this value proactively.

query if pin code activated

```
COE
```

COE[:{ONOFF}]

turn on/off pin code for bluetooth connection, device will reboot when execute this command

set connection code to 1234

```
COD:1234
```

COD[:{PIN}]

set/get pin code for bluetooth connection, default



Playback

SUMMARY

Command	Ver	Response	Ctrl
SRC[:{source}]	3	SRC:{source}	*
POP	3		*
STP	3		*
NXT	3		*
PRE	3		*
PST:{preset}	3		*
LPM[:{loopmode}]	3	LPM[:{loopmode}]	*
BTC[:{onoff}]	3		*
PLA	3	PLA:{onoff}	

Command	Ver	Response	Ctrl
CHN	3	CHN:{channel}	
MRM	3	MRM:{mrmmode}	
TIT	4	TIT:{hextext}	
ART	4	ART:{hextext}	
ALB	4	ALB:{hextext}	
VND	4	VND:{vendor}	
ELP	4	ELP:{elapsed}	
PLI	4	PLI:{playlist_info}	
APL[:{onoff}]	5	APL:{onoff}	*

switch source to bluetooth

SRC:BT

SRC[:{SOURCE}]

get or set input source of device

{source}	description
NET	network
BT	bluetooth
USBDAC	USB DAC
LINE-IN	line-in
OPT	Optical
COAX	Coaxial
LINE-IN2	extra line-in
OPT2	extra Optical
COAX2	extra Coaxial
HDMI	HDMI ARC

POP

play or pause, available in network playback and bluetooth

STP

stop, available only in network playback

next track

NXT

NXT

next track, available in network playback and bluetooth

PRE

previous track, available in network playback and bluetooth

play preset #1

PST:1

PST:{PRESET}

start to play preset playlist

LPM[:{LOOPMODE}]

set/get loop and shuffle mode, available in network playback.

{loopmode}	description
REPEATALL	repeat all in playlist
REPEATONE	repeat track
REPEATSHUFFLE	repeat all and shuffle
SHUFFLE	shuffle and stop when all tracks played
SEQUENCE	stop when reach end of playlist

close current bluetooth connection

BTC:0

BTC[:{ONOFF}]

disconnect/reconnect bluetooth

PLA

network playing state

CHN

query multiroom channel status, does not support to set from API

{channel}	description
S	stereo mode
L	left channel
R	right channel

MRM

query multiroom mode, does not support to set from API

{mrmmode}	description
S	multiroom slave

{mrrmmode}	description
M	multiroom master
N	normal mode

TIT

notification messages for song metadata title. (not work when sent in TCP)

ART

notification messages for song metadata artist. (not work when sent in TCP)

ALB

notification messages for song metadata album. (not work when sent in TCP)

VND

notification messages for song metadata vendor. (not work when sent in TCP)

{vendor} will have the following value:

- spotify
- qplay
- dlna
- airplay
- upnp
- phone
- usb
- tidal
- napster
- qobuz
- amazon
- tunein
- iheart
- vtuner
- http
- other

ELP

notification messages for elapsed and duration of current track. {elapsed} will have unit ms, and a sample: 31251/212000

(not work when sent in TCP)

message received when query playlist info

PLI:1/23

PLI

query current track index and number of playlist, {playlist_info} will be in this format index/count, and index is start from 1. eg: 1/23 means now playing the first song in playlist contains 23 songs in total.

APL[:{ONOFF}]

set/get autoplay feature. When enabled, the device will try to play last playlist on boot.

Audio

SUMMARY

Command	Ver	Response	Ctrl
AUD[:{onoff}]	3	AUD:{onoff}	*
VOL[:{volume}]	3	VOL:{volume}	*
MUT[:{onoff}]	3	MUT:{onoff}	*
BAS[:{tone}]	3	BAS:{tone}	*
TRE[:{tone}]	3	TRE:{tone}	*
MID[:{tone}]	6	MID:{tone}	*

Command	Ver	Response	Ctrl
VBS[:{onoff}]	3	VBS:{onoff}	*
BAL[:{balance}]	5	BAL:{balance}	*
VOF[:{volume}]	5	VOF:{volume}	*
VOG[:{volume}]	6	VOG:{volume}	*
PEQ	6	PEQ:{eqlist}	
EQS[:{eqidx}]	6	EQS:{eqidx}	*
VST[:{step}]	6	VST:{step}	*
EQE[:{onoff}]	7	EQE:{onoff}	*
CFE[:{onoff}]	7	CFE:{onoff}	*
CFF[:{cffreq}]	7	CFF:{cffreq}	*

AUD[:{ONOFF}]

set/get audio output

set system volume to 50

```
VOL:50
```

VOL[:{VOLUME}]

set/get master volume

MUT[:{ONOFF}]

set/get mute

set bass to +2

```
BAS:2
```

BAS[:{TONE}]

set/get bass tone, and {tone} ranges [-10,10], unit dB

TRE[:{TONE}]

set/get treble tone

MID[:{TONE}]

set/get middle tone

enable virtual bass

```
VBS:1
```

VBS[:{ONOFF}]

set/get virtual bass

set the sound balance to the right side

BAL:50

BAL[:{BALANCE}]

set/get audio output balance, the {balance} ranges [-100,100], available only when output mode is stereo. Will attenuate the left channel output when {balance} is greater than 0, and attenuate the right channel output when {balance} is less than 0.

VOF[:{VOLUME}]

set/get fixed volume output, the {volume} ranges [0,100], and will be disabled when set to 0. And if set to non zero value, the system will have fixed output level, and can't be adjusted by volume control.

VOG[:{VOLUME}]

set volume of grouped playback (multiroom)
the query result of EQ groups

PEQ:0@Flat,1@Classical,2@Pop,3@Jazz,4@Rock,5@Vocal

PEQ

query system EQ group list, the format of {eqlist} will be index@name, and seperated by ,.
select EQ group #1

EQS:1

EQS[:{EQIDX}]

get/set EQ group

VST[:{STEP}]

set/get volume step, {step} ranges [0,10]. It will affect the volume value changes when press once vol+ or vol- button.

EQE[:{ONOFF}]

enable/disable EQ

CFE[:{ONOFF}]

enable/disable crossfilter, when activated the stereo output will have a high-pass filter. and the DAC-X output will have a low-pass filter.

CFF[:{CFFREQ}]

set/get crossfilter frequency point, {cffreq} ranges [50,300]

Misc

SUMMARY

Command	Ver	Response	Ctrl
VER	3	VER:{version}	
LED[:{onoff}]	3	LED:{onoff}	*
BEP[:{onoff}]	3	BEP:{onoff}	*
PMT[:{onoff}]	4	PMT:{onoff}	*

Command	Ver	Response	Ctrl
DLY[:{mute_delay}]	4	DLY:{mute_delay}	*
MXV[:{volume}]	4	MXV:{volume}	*
ASW[:{onoff}]	4	ASW:{onoff}	*
POM[:{source}]	4	POM:{source}	*
VOS:{onoff}	4	VOS:{onoff}	*
LST	7	LST:{sources}	
SOP:{onoff}	7	SOP:{onoff}	*

response of current version

VER:44-c7c30da5-8

VER

firmware version, and the {version} will contain the version number, short git commit, and API level, connected with .

turn off beep sound

BEP:0

BEP[:{ONOFF}]

Query or turn on/off beep sound for key press

LED[:{ONOFF}]

Query or turn on/off led or display

PMT[:{ONOFF}]

Query or turn on/off the prompt voice, device will reboot on this command

ASW[:{ONOFF}]

Query or turn on/off auto switch mode. When enabled, will try return previous input source when connection lost. eg: will switch to previous input when plug-out detected in AUX mode.

VOS:{ONOFF}

Query or turn on/off volume sync feature. When enabled, and adjusting volume on master device (multiroom group) with remote or button, will sync the changes to slave devices.

SOP:{ONOFF}

Query or turn on/off standby on power. When enabled, device will enter standby mode on power supplied.

set max volume to 80

MXV:80

MXV[:{MAX_VOLUME}]

set/get max volume, {max_volume} ranges [30,100]

DLY[:{MUTE_DELAY}]

set/get delay time to auto mute when detected no audio input. {mute_delay} ranges [0,32767]

supported input sources of Up2Stream Pro

```
LST:NET,BT,LINE-IN,USBDAC
```

LST

query system available input sources

POM[:{SOURCE}]

set/get power on mode, when set an available source, the device will select the specified source on power up.

4 Zone Model

We've produced 4 models with 4 zone, MA400/HA400, and M400/H400 which without contain internal amplifier. There's a master MCU to manage the 4 zones on these models. And the RS232 port is connect to this master MCU, and it could transfer commands to specific zone. We defined a special command **ZON** to redirect commands sent to specific zones. Each zone has an id which is used to identify zone physically. The param used to **ZON** is logic zone id. The logic zone id is by default the same with physical zone id, and could be changed with command **IDS**.

This RS232 managing port support to connect in casacading, that means you could use a host to control 2 or more MA400s. And before connect together, you need to assign different logic zone id first. So you're able to identify each zone.

zone manage

set volume of zone 1 to 50

```
ZON:1:VOL:50;
```

ZON:{LOGICZONEID}:{MSG}

Redirect {msg} to specific zone with {logiczoneidx}, {msg} is the UART command and parameter defined previously. And response will also be redirected in the same format.

set logic id of zone 1 to 5

```
IDS:1:5;
```

IDS query result

```
IDS:5,2,3,4;
```

IDS[:{ZONEID}]{LOGICZONEID}

query or set/get logic zone id, the query response will contain the logic zone id of all zones. The logic zone id ranges [1,127].

Default Configuration

For the Up2Stream series, users are building various products based on it. So we defined a set of API to set the default settings of these boards. eg: default device name, disable the beep sound by default, etc.

example of set default volume to 30

DEF:VOL:30

The default configuration will be sent as param of API DEF, and is defined as sub level API.

Sub API

All sub APIs is supposed to send with DEF.

SUMMARY

Command	Ver	Response
LTP[:{led_type}]	6	LTP:{led_type}
NAM[:{hextext}]	6	NAM:{hextext}
FXN[:{onoff}]	6	FXN:{onoff}
VOL[:{volume}]	6	VOL:{volume}
VBS[:{onoff}]	6	VBS:{onoff}
PMT[:{onoff}]	6	PMT:{onoff}
VOS[:{onoff}]	6	VOS:{onoff}
BEP[:{onoff}]	6	BEP:{onoff}
MDL[:{hextext}]	6	MDL:{hextext}
VST[:{step}]	6	VST:{step}
SAV	6	SAV:1
SEN[:{source}={onoff}]	7	SEN:{sources} on query, SEN:1 on set
POM[:{source}]	7	POM:{source}
MXV[:{max_volume}]	7	MXV:{max_volume}
COE[:{onoff}]	8	COE:{onoff}
COD[:{pin}]	8	COD:{pin}
LAP[:{onoff}]	8	LAP:{onoff}

set to use single PIN to drive LED

DEF:LTP:PIN

LTP[:{LED_TYPE}]

Change LED driving method, availabe only on model Up2Stream PRO and AMP2.0.

{led_type}	desc
UND	firmware default
RGB	working in RGB mode
PIN	driven with single PIN, and some modes are sharing PIN.

NAM[:{HEXTEXT}]

Set default device name

FXN[:{ONOFF}]

Set whether to restore the default device name when do factory reset. It's off by default

VOL[:{VOLUME}]

Set default volume level

turn on virtual bass by default

```
DEF:VBS:1
```

VBS[:{ONOFF}]

Set default virtual bass state, default off.

PMT[:{ONOFF}]

Set default prompting voice state, default on.

VOS[:{ONOFF}]

Set default volume sync state, default off.

BEP[:{ONOFF}]

Set default beep sound state, default on.

MDL[:{HEXTEXT}]

Set model name, it's not used in device actually, could be used to identify your model.

VST[:{STEP}]

Set default volume step, default 3.

SAV

Save the settings. And this messages has to be sent in the last to make sure the changes will take effect. And after configured, you need to restore factory default to verify.

disable LINE-IN input

```
DEF:SEN:LINE-IN=0
```

SEN query response

```
DEF:SEN:NET,BT,USB DAC
```

SEN[:{SOURCE}]{ONOFF}]

Disable/enable some input modes. You could only enable the existing input modes supported by model.

And this command is a bit special, it will just save and restore factory default when settings updated.

POM[:{SOURCE}]

Set the default input method when power on.

MXV[:{MAX_VOLUME}]

Set the default max volume, default 100.

COE[:{ONOFF}]

Set the default bluetooth pin code protection state, default off.

COD[:{PIN}]

Set the default bluetooth pin code, default 0000.

LAP[:{ONOFF}]

Set the default auto play feature of AXX module, default off.